# HW 3 — Escape from a Dielectric & Jackson 7.12

black = write it down · red = the answer · teal = my note, do not copy · grey box = optional

## Problem 1 — Escape from a dielectric

> A point source of light is embedded near the flat surface of a dielectric of index $n$. Treat the emitted light as plane waves (rays) going out **isotropically**. Find the **fraction** of rays that can refract out into the vacuum above.

ANSWER

$$f = \frac{1}{2}\left[1 - \sqrt{1 - 1/n^2}\right]$$

*Glass ($n = 1.5$) gives **12.7%**. For big $n$ it goes as $1/4n^2$.*

### Step 1 — Which rays escape?

Let $\theta$ be the angle a ray makes with the surface normal — this is its angle of incidence.

**Snell's law** gives:

$$n \sin\theta = n' \sin r$$ Lec 4, slide 7

Above is vacuum, so $n' = 1$:

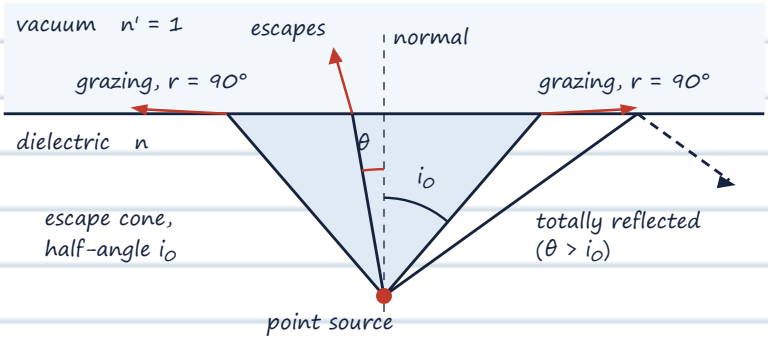
$$\sin r = n \sin\theta$$

But $\sin r$ cannot exceed 1. So a refracted ray exists only if

$$n \sin\theta \le 1, \quad \text{that is} \quad \theta < i_o, \quad \text{where} \quad \sin i_o = 1/n$$

If $\theta > i_o$, $\sin r$ would have to exceed 1 — impossible. No refracted ray exists, so the wave is **totally internally reflected.** Lec 4, slide 8

So only rays aimed steeply upward escape. Together they fill the **escape cone** — a cone of **half-angle $i_o$** about the normal. Rays aimed downward never reach the surface.




*θ = angle of any ray from the normal. $i_O$ = the **critical angle**, the largest θ that still escapes.*

*__Grazing__: at $\theta = i_O$, $\sin r = n \sin i_O = 1$, so $r = 90°$ — the refracted ray emerges along the surface.*

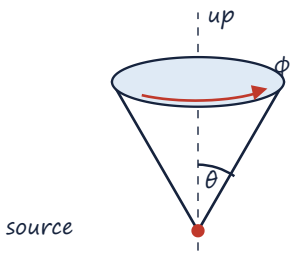
## Step 2 — Solid angle of the escape cone

The source shines the same in every direction. So the fraction of light that escapes is just the fraction of **directions** that escape.

To name a direction in 3D you need **two** angles:

- **θ = polar angle** — how far you tilt from the normal *(like a slope angle)*
- **ϕ = azimuthal angle** — the rotation about the normal *(like a compass bearing)*

*The picture above is a **flat slice** — it shows one ϕ only. Spin that whole picture around the dashed normal line and the triangle sweeps out the real 3D cone. That spin is ϕ.*




*θ opens the cone; ϕ runs around its rim.*

A small patch of directions has height $d\theta$ and width $\sin\theta\, d\phi$, so its **solid angle** is

$$d\Omega = \sin\theta\, d\theta\, d\phi$$

*Why $\sin\theta$? Circles near the top of a sphere are small. At $\theta = 0$ the circle is a point ($\sin 0 = 0$); at $\theta = 90°$ it is the widest circle ($\sin 90° = 1$).*

Integrate over every direction in the cone: $\theta$ from 0 to $i_o$, and $\phi$ all the way round from 0 to $2\pi$, since the cone looks the same from every side:

$$\Omega_{escape} = \int_0^{2\pi} d\phi \int_0^{i_o} \sin\theta\, d\theta = 2\pi\,[-\cos\theta]_0^{i_o} = 2\pi(1 - \cos i_o)$$

## Step 3 — Convert to a fraction of all directions

All directions together make $4\pi$ steradians, so the escaping fraction is

$$f = \frac{\Omega_{escape}}{4\pi} = \frac{2\pi(1 - \cos i_o)}{4\pi} = \frac{1 - \cos i_o}{2}$$

## Step 4 — Express the answer in terms of n

From Step 1, $\sin i_o = 1/n$. Using $\sin^2 + \cos^2 = 1$:

$$\cos i_o = \sqrt{1 - \sin^2 i_o} = \sqrt{1 - 1/n^2}$$

Substitute into Step 3:

FINAL ANSWER

$$f = \frac{1}{2}\left[\,1 - \sqrt{1 - 1/n^2}\,\right]$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

OPTIONAL **Checks.** $n \to 1$ gives $f \to ½$ (no index step, so everything going up escapes) ✓ big n gives $f \approx 1/4n^2$ ✓ water 17.0% | glass 12.7% | diamond 4.5% | GaAs 2.1% — which is why LEDs need domes.

# Problem 2 — Jackson 7.12

> The time dependence of electrical disturbances in good conductors is governed by the frequency-dependent conductivity (7.58). Consider **longitudinal** electric fields in a conductor, using Ohm's law, the continuity equation, and the differential form of Coulomb's law.
> **(a)** Show that the time-Fourier-transformed charge density satisfies $[\sigma(\omega) - i\omega\varepsilon_0]\rho(\mathbf{x},\omega) = 0$.
> **(b)** Using $\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$ with $\sigma_0 = \varepsilon_0\omega_p^2\tau$, show that for $\omega_p\tau \gg 1$ any initial disturbance oscillates at the plasma frequency and decays with constant $\lambda = 1/2\tau$.

**ANSWER**

**(a)** $[\sigma(\omega) - i\omega\varepsilon_0]\,\rho(\mathbf{x},\omega) = 0$

**(b)** $\omega^2 + i\omega/\tau - \omega_p^2 = 0 \;\Rightarrow\; \omega \approx \pm\omega_p - \dfrac{i}{2\tau} \;\Rightarrow\; \rho \propto e^{\mp i\omega_p t}\, e^{-t/2\tau}$

*— it **oscillates** at $\omega_p$ and **decays** with $\lambda = 1/2\tau$.*

*Every step below is demanded by a phrase in the question. Nothing is invented:*

| | |
|---|---|
| *"using Ohm's law, the continuity equation, and the differential form of Coulomb's law"* | **Step 0** |
| *"the **frequency-dependent** conductivity $\sigma(\omega)$"* — only meaningful one $\omega$ at a time | **Step 1** |
| *"the **time-Fourier-transformed** charge density"* | **Step 1** |
| *the target holds only $\rho(\mathbf{x},\omega)$ — no $\vec{J}$, no $\vec{E}$* | **Step 2** |
| *"**using the representation** $\sigma(\omega) = \sigma_0/(1-i\omega\tau)$, **where** $\sigma_0 = \varepsilon_0\omega_p^2\tau$"* | **Step 3** |
| *nothing — this part is just algebra* | **Step 4** |
| *"**in the approximation** $\omega_p\tau \gg 1$"* | **Step 5** |

*Step 0 sets up both parts. **Steps 1–2 answer part (a); Steps 3–5 answer part (b).***

## Step 0 — The laws we use

*The problem tells us to use the first three.*

$\vec{J} = \sigma(\omega)\,\vec{E}$ *(Ohm — current follows the field)*

$\nabla\cdot\vec{J} + \partial\rho/\partial t = 0$ *(continuity — charge is conserved)*

$\nabla\cdot\vec{E} = \rho/\varepsilon_0$ *(Coulomb, differential form)*

$\omega_p^2 = \dfrac{Ne^2}{\varepsilon_0 m}$ *(plasma frequency)* Lec 6

***Notation.*** *He writes $\sigma_0 = Ne^2/(m\gamma_0)$; Jackson writes $\sigma_0 = \varepsilon_0\omega_p^2\tau = Ne^2\tau/m$. Same thing, so $\tau = 1/\gamma_0$*

## PART (a) — show that $[\,\sigma(\omega) - i\omega\varepsilon_0\,]\,\rho(\mathbf{x},\omega) = 0$

### Step 1 — Fourier transform in time

The continuity equation contains $\partial\rho/\partial t$. Part (b) asks which frequency the charge settles at, and a frequency cannot be read off a differential equation — only off an algebraic one. So the derivative has to go.

Assume every quantity oscillates at a single frequency $\omega$:

$$f(\mathbf{x},t) = f(\mathbf{x},\omega)\,e^{-i\omega t} \qquad f \text{ stands for } \rho,\ \vec{J} \text{ or } \vec{E}$$

$f(\mathbf{x},\omega)$ is the **amplitude**: it depends on position only, and all of the time dependence sits in $e^{-i\omega t}$.

Differentiating in time therefore acts on the exponential alone:

$$\frac{\partial \rho}{\partial t} = \rho(\mathbf{x},\omega)\,(-i\omega)\,e^{-i\omega t} = -i\omega\,\rho \qquad \text{since } \rho(\mathbf{x},\omega)e^{-i\omega t} \text{ is } \rho \text{ itself}$$

That is, **a time derivative is equivalent to multiplication by $-i\omega$**: $\partial/\partial t \to -i\omega$

Substituting into the continuity equation, every term carries $e^{-i\omega t}$, which is never zero and therefore cancels:

$$\nabla\cdot\vec{J}(\mathbf{x},\omega) - i\omega\rho(\mathbf{x},\omega) = 0$$

> "Fourier transform in time" is the formal name for this substitution — and it is the phrase the question itself uses when it asks for the time-Fourier-transformed charge density.

## Step 2 — Substitute Ohm's law, then Coulomb's law → part (a)

We want an equation in $\rho$ alone. Substitute $\vec{J} = \sigma(\omega)\vec{E}$:

$$\sigma(\omega)\,\nabla\cdot\vec{E} - i\omega\rho = 0$$

Now substitute $\nabla\cdot\vec{E} = \rho/\varepsilon_0$:

$$\frac{\sigma(\omega)}{\varepsilon_0}\,\rho - i\omega\rho = 0$$

Multiply by $\varepsilon_0$ and factor out $\rho$:

**FINAL ANSWER — PART (a)**

$$\left[\,\sigma(\omega) - i\omega\varepsilon_0\,\right]\rho(\mathbf{x},\omega) = 0$$

Divide by $(-i\omega)$ and this is $[\varepsilon_0 + i\sigma/\omega]\rho = 0$ — his conductor permittivity set to zero. (Lec 7, slide 2)

## PART (b) — show it oscillates at $\omega_p$ and decays with $\lambda = 1/2\tau$

### Step 3 — Take part (a)'s equation and put in the given $\sigma(\omega)$

Part (b) says "**using the representation** $\sigma(\omega) = \sigma_o/(1-i\omega\tau)$". Using it **where?** — in the equation part (a) has just produced. So start by writing that down again:

$$[\ \sigma(\omega) - i\omega\varepsilon_o\ ]\,\rho(\mathbf{x},\omega) = 0$$ *(the answer to part (a))*

A real disturbance has $\rho \neq 0$, so the bracket itself must vanish:

$$\sigma(\omega) = i\omega\varepsilon_o$$

Substitute the given $\sigma(\omega) = \sigma_o/(1-i\omega\tau)$ and clear the fraction:

$$\sigma_o = i\omega\varepsilon_o(1 - i\omega\tau) = i\omega\varepsilon_o + \omega^2\varepsilon_o\tau$$

Substitute the given $\sigma_o = \varepsilon_o\omega_p{}^2\tau$, then divide by $\varepsilon_o\tau$:

$$\boxed{\omega^2 + \frac{i\omega}{\tau} - \omega_p{}^2 = 0}$$

### Step 4 — Solve the quadratic

This is a quadratic in $\omega$. With $a = 1$, $b = i/\tau$, $c = -\omega_p{}^2$, and $(i/\tau)^2 = -1/\tau^2$:

$$\omega = -\frac{i}{2\tau} \pm \sqrt{\omega_p{}^2 - 1/4\tau^2}$$

### Step 5 — Use $\omega_p\tau \gg 1$

Since $\omega_p\tau \gg 1$, the term $1/4\tau^2$ is negligible beside $\omega_p{}^2$, so the square root is just $\omega_p$:

$$\omega \approx \pm\omega_p - \frac{i}{2\tau}$$

Since $\rho(t) \propto e^{-i\omega t}$, put this $\omega$ back in:

$$\rho(t) \propto e^{\mp i\omega_p t} \cdot e^{-t/2\tau}$$

The **real** part of $\omega$ gives the oscillation; the **imaginary** part gives the decay.

**FINAL ANSWER — PART (b)**

$$\text{oscillates at } \omega_p\ , \quad \text{decays with } \lambda = \frac{1}{2\tau}$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

## EXTRA — not required for the answer

OPTIONAL **Why you must not use the DC value $\sigma_o$.** It gives $\omega = -i\sigma_o/\varepsilon_o$, which is purely imaginary — so **no oscillation at all**, and a decay $\lambda_w = \sigma_o/\varepsilon_o = \omega_p^2\tau$. For $\omega_p\tau = 100$ that is 20000× too fast — Jackson's "(wrong) decay constant". The disturbance happens at $\omega \approx \omega_p$, where $\omega\tau$ is huge, so the DC value simply does not apply.

OPTIONAL **The decay is exact.** The imaginary part of both roots is *exactly* $-1/2\tau$ whenever $\omega_p\tau > ½$; only the oscillation frequency is approximated. For $\omega_p\tau = 100$ the exact roots are $\pm 99.9987/\tau - 0.5i/\tau$.